Unified chassis control: a model-free approach

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Abstract

Automotive safety has been a critical focus for both national and international research institutions in recent years. The combined control of lateral dynamics and roll stability in vehicles has been extensively studied, often relying on automotive models to design and tune controllers. However, these models frequently include parametric uncertainties, nonlinear dynamics, and simplifying assumptions, which can lead to suboptimal controller performance in real-world applications. This work proposes a unified chassis controller (UCC) to regulate lateral and roll dynamics during abrupt maneuvers and evaluates a model-free control (MFC) strategy against representative benchmark controllers. The comparison includes a fixed-parameter 2PID+fuzzy controller, a gain-scheduled linear quadratic regulator (LQR), and the built-in CarSim® ESC baseline, all assessed within the same co-simulation environment combining MATLAB/Simulink® and CarSim®. Results demonstrate that while the benchmark controllers degrade under varying road conditions, the MFC-based UCC achieves superior robustness and performance, effectively regulating the state variables across different scenarios. These findings highlight the robustness and adaptability of MFCs, making them a promising solution for enhancing vehicle safety.

Keywords: UCC, robustness, CarSim®, MFC

1 Introduction

Road traffic accidents remain a major global concern, causing approximately 1.3 million deaths and tens of millions of injuries each year (Naciones Unidas (Noticias ONU), 2022). Many of these incidents are associated with loss of vehicle stability during abrupt maneuvers, tire failures, or adverse road conditions (Friedman et al., 2010). Such situations are typically characterized by excessive side-slip angle and yaw rate, which can lead to oversteer or understeer, as well as increased rollover risk due to significant roll angle variations (Stolle et al., 2011; Zhan, 2017; Nguyen, 2023). Consequently, side-slip angle, yaw rate, and roll angle have been widely recognized as the key state

variables for vehicle stability monitoring and control, serving as the primary feedback signals in active safety systems (Rajamani, 2012).

To address these challenges, modern vehicles incorporate advanced control systems such as Electronic Stability Control (ESC), which improves lateral stability through selective braking and torque modulation (Aleksander and Edward, 2007). In parallel, suspension-based systems, including active and semi-active suspensions, are used to regulate vertical dynamics and enhance ride comfort.

Unified Chassis Control (UCC) extends these concepts by coordinating multiple subsystems to simultaneously regulate lateral and vertical vehicle dynamics, improving overall stability and safety, particularly under extreme driving conditions (Wanki et al., 2012). However, achieving this coordination is non-trivial, as lateral stability enhancement, rollover mitigation, and actuator allocation are inherently competing objectives, requiring careful trade-off management to avoid control conflicts between subsystems (Falcone et al., 2007).

Most existing UCC strategies rely on model-based control approaches, which can be broadly categorized by their underlying vehicle representation. On one end, simplified formulations such as the bicycle model have been widely adopted within Model Predictive Control (MPC) frameworks, offering computational tractability for real-time path-following and yaw stabilization tasks (Ahmadi, 2021; Gidlewski et al., 2017; Yakub et al., 2016). On the other end, multi-degree-of-freedom formulations have been proposed to capture the coupling between lateral, vertical, and suspension dynamics more faithfully: Lai and Jiang (2021) developed an 18-DOF chassis model that enables high-fidelity simulation of cross-subsystem interactions, while Chen et al. (2021) extended this perspective to electric vehicles by explicitly incorporating wheel vertical vibrations into a UCC architecture.

Although these high-fidelity representations improve prediction accuracy and allow for a more principled treatment of lateral-vertical coupling, they introduce substantial computational overhead and exhibit strong sensitivity to parametric uncertainty, limiting their portability across varying road conditions and vehicle configurations (Guo et al., 2020; Lew et al., 2024).

To mitigate these limitations, several works have introduced uncertainty compensation mechanisms based on data-driven or adaptive techniques. Neural network-based controllers have been widely used to approximate unknown nonlinearities and disturbances in vehicle dynamics, enhancing robustness under varying conditions. Similarly, fuzzy and neuro-fuzzy approaches enable online adaptation of control parameters, while sliding mode control combined with intelligent approximators provides robustness against disturbances and model mismatch (Swain et al., 2021; Guo et al., 2021). More recently, hybrid strategies combining model-based controllers with learning-based compensation terms have demonstrated improved performance in the presence of uncertainties (Levine et al., 2016).

In this context, model-free control (MFC) has emerged as a structurally simpler alternative that reduces dependence on an explicit vehicle model; however, its performance within a UCC architecture has only rarely been assessed against representative baselines (Sánchez-Morales et al., 2023; Mazzilli et al., 2021). In particular, many comparative studies in this domain rely primarily on simplified fixed-parameter benchmarks, often limited to lateral PID-based ESC schemes, leaving open the question of how MFC compares with more meaningful references such as gain-scheduled controllers and simulator-embedded baseline strategies under consistent evaluation conditions.

To address this gap, this paper proposes a model-free UCC strategy for the simultaneous regulation of side-slip angle, yaw rate, and roll angle, and evaluates it in a MATLAB/Simulink-CarSim® co-simulation environment under a sine-with-dwell (SWD) maneuver with different road friction coefficients. The main contributions of this work are threefold: first, the design and implementation of an MFC-based UCC architecture for the joint regulation of lateral and vertical vehicle dynamics without relying on an explicit vehicle model; second, a comparative assessment against representative benchmark controllers, including a fixed-parameter 2PID+fuzzy controller, a gain-scheduled LQR, and a CarSim® built-in baseline controller; and third, an analysis of practical closed-loop stability, robustness to friction

variation, and control performance limits of the proposed approach.

The remainder of the paper is organized as follows. Section 2 formulates the control problem, including the regulated variables, references, actuator constraints, and maneuver evaluation criteria. Section 3 presents the benchmark controllers and the comparative framework. Section 4 describes the proposed MFC-based UCC architecture, including the ultra-local model, estimators, control allocation, and parameter selection. Section 5 details the experimental design, Section 6 discusses the simulation results, and Section 7 concludes the paper.

2 Control Problem Formulation

This section formally defines the control problem addressed in this work. The regulated variables, reference generation strategy, actuator constraints, and maneuver evaluation criteria are described, establishing the framework within which the proposed UCC is designed and assessed.

2.1 Vehicle States and Control Objectives

The vehicle dynamics relevant to this work are characterized by three state variables: the side-slip angle β, the yaw rate $\dot{\psi}$, and the roll angle ϕ. Together, these variables capture the coupled lateral and vertical behavior of the chassis during abrupt maneuvers and serve as the primary feedback signals for the proposed UCC.

The side-slip angle is defined as:

$$\beta = \tan^{-1}\left(\frac{V_y}{V_x}\right), \tag{1}$$

where V_y and V_x are the lateral and longitudinal vehicle velocities, respectively. In the present co-simulation setup, both V_x and V_y are directly obtained from CarSim®. Accordingly, β is computed from ideal state information available in simulation, rather than being reconstructed from measured signals through an estimation algorithm. This choice is suitable for comparative controller assessment, although it abstracts from the state-estimation problem that would arise in a real-vehicle implementation.

The reference for β is set to zero, reflecting the control objective of minimizing lateral drift. The controller acts only when the side-slip angle exceeds the safety threshold:

$$|\beta| > \beta_{\text{lim}} = 2^\circ. \tag{2}$$

The yaw rate reference $\dot{\psi}_D$ is computed dynamically based on longitudinal speed, front wheel steering angle, and road friction coefficient, as described in Section 2.2. The roll angle reference is always set to zero, as the control objective is to minimize chassis tilt throughout the maneuver.

For the sine-with-dwell (SWD) maneuver considered in this study, choosing $\phi_{\text{ref}} = 0$ is appropriate because the main control objective is to minimize body roll and preserve rollover safety margins during a severe transient maneuver, rather than to track a nonzero equilibrium roll angle. From this perspective, a zero roll reference represents the nominal upright posture of the vehicle and allows the controller to directly counteract excessive lateral load transfer.

2.2 Reference Generation and Operating Constraints

The desired yaw rate is computed as:

$$\dot{\psi}_D = \min\left(\left|\frac{\mu g}{V_x}\right|, \left|G_{\dot{\psi}}\,\delta_f\right|\right) \cdot \text{sign}(\delta_f), \tag{3}$$

where:

$$G_{\dot{\psi}} = V_x/(L + K_{us}V_x^2) \tag{4}$$

is the yaw rate gain, L is the vehicle wheelbase, K_{us} is the understeer coefficient, δ_f is the front wheel steering angle, μ is the road friction coefficient, and g is the gravitational acceleration. The parameters required to compute $G_{\dot{\psi}}$ are taken from Arronte et al. (2023). The same reference provides the full set of vehicle parameters used in the CarSim® model considered in this work.

In this study, the road friction coefficient (μ) is imposed as a simulation parameter and kept constant within each test scenario. Therefore, μ is not estimated online, but prescribed a priori in CarSim® to define the operating condition under which the controllers are evaluated. This choice is consistent with the main objective of

the experiments, namely, to assess controller performance and robustness under different known friction levels.

2.3 Control Inputs and Actuator Constraints

The high-level controller produces two virtual control inputs: a yaw moment M_z, acting about the vehicle's vertical axis to regulate lateral dynamics, and a roll moment M_x, acting about the longitudinal axis to regulate vertical dynamics. These moments are subject to the following saturation constraints:

$$M_z \in [-8000,\ 8000]\ \text{Nm}, \tag{5}$$

$$M_x \in [-1500,\ 1500]\ \text{Nm}. \tag{6}$$

The saturation limits were set in accordance with the maximum torque capacity of the anti-lock braking system (ABS) and active anti-roll bar (AARB) subsystems as defined in the CarSim® SUV model used throughout this work, consistent with the simulation framework established in (Arronte et al., 2023).

At the actuation level, M_z is decomposed into individual wheel brake pressure commands by the ABS subsystem, while M_x is translated into active anti-roll bar (AARB) force commands. Both decompositions are handled internally by CarSim®, as described in Section 4.3.

2.4 Maneuver Completion and Failure Criteria

The performance of the UCC is evaluated under a sine-with-dwell (SWD) maneuver, a standardized test used to assess vehicle response to sudden evasive inputs, as specified in FMVSS 126 and UN/ECE Regulation No. 13-H (Mechanical Simulation Corporation, 2012). A maneuver is considered successfully completed if the vehicle maintains controlled motion throughout the full duration of the steering input without loss of stability.

In the context of the SWD maneuver, maneuver failure is determined following the criteria established by FMVSS 126 and implemented in CarSim® (Mechanical Simulation Corporation, 2021): the vehicle is considered to have failed the maneuver if the yaw rate measured at 1.0 s after the completion of steering exceeds 35% of the peak yaw rate recorded during the maneuver, or if the yaw rate at 1.75 s after the completion of steering exceeds 20% of that peak value (Mechanical Simulation Corporation, 2012). In practice, within the CarSim® co-simulation environment, this condition manifests as an abrupt termination of the simulation due to vehicle loss of control, which is the failure indicator used throughout Section 6.

In addition to maneuver completion, controller performance is quantified using the root mean square (RMS) of each regulated variable, the root mean square error (RMSE) with respect to the corresponding reference, and the maximum absolute value (MAX) recorded during the maneuver. Control effort is monitored through the time histories of M_z and M_x, with particular attention to saturation occurrence and duration.

3 Benchmark Controllers and Comparative Framework

To provide a rigorous and meaningful assessment of the proposed MFC-based UCC, three benchmark controllers are considered. Each controller operates within the same UCC architecture described in Section 4.3, receives the same input signals, and is subject to the same actuator saturation constraints defined in Section 2.3. The benchmarks are selected to represent progressively more capable reference strategies, allowing the comparison to go beyond the fixed-gain baselines that are common in the literature (Sánchez-Morales et al., 2023).

3.1 Fixed-Parameter 2PID+Fuzzy Controller

The first benchmark is the fixed-parameter UCC architecture reported in (Arronte Delgado, 2024). In that design, the lateral dynamics are regulated by two independent PID controllers with hysteresis acting on the side-slip angle β and the yaw rate $\dot{\psi}$. Their outputs are combined into a corrective yaw moment M_z, which is implemented by differential braking in the ESC layer. As in the earlier ESC design, the lateral PID branches are only activated when the corresponding regulated variables leave prescribed admissible bands, which reduces unnecessary braking effort during mild transients.

The reference described in (Arronte Delgado, 2024) forms an integrated "2PID+fuzzy" structure. The vertical dynamics are handled by a Mamdani-type fuzzy controller that uses the roll angle ϕ and the roll rate $\dot{\phi}$ as inputs and generates the corrective roll moment M_x as output. In addition, the fuzzy layer is complemented by a feedforward stage intended to attenuate the effect of lateral-force disturbances on the roll dynamics.

Accordingly, this benchmark acts on the same two control moments considered in the present UCC framework, namely M_z for lateral stabilization and M_x for roll mitigation. It therefore provides a fixed-parameter reference for the joint lateral-roll regulation problem addressed in this paper, while retaining a comparatively simple and interpretable control structure.

3.2 Gain-Scheduled LQR Controller

The second benchmark is a gain-scheduled linear quadratic regulator (LQR), in which the state-feedback gain matrix is adapted as a function of the longitudinal vehicle speed V_x.

The LQR benchmark is derived from the bicycle-model framework described in (Arronte et al., 2023), but extended to represent the coupled lateral and roll dynamics required by the present UCC problem. In particular, the state-space model is written in the descriptor form $E\dot{x} + Fx = Gu$, with state vector $x = [v\ \dot{\psi}\ \dot{\phi}\ \phi]^\top$ and input vector $u = [\delta\ M_z\ M_x]^\top$, where δ is the steering input, M_z is the external yaw moment, and M_x is the external roll moment. Relative to the model used in (Arronte et al., 2023), the main modification lies in the input matrix G, which is reformulated to account simultaneously for steering excitation and for the two control moments used by the UCC architecture. This yields a speed-dependent linear model that preserves the bicycle-model interpretation while explicitly capturing the actuation channels required for both lateral stabilization and roll regulation.

The scheduling structure is defined over 11 longitudinal-speed operating points, from 20 km/h to 120 km/h in increments of 10 km/h. For each point, a state-feedback gain matrix is precomputed, and the gains used during simulation are obtained by interpolating these matrices in MATLAB using the `interp1` function as a function of the measured vehicle speed V_x. The resulting gain-scheduling mechanism is applied to the complete controlled system, so that both the lateral dynamics and the roll dynamics are governed by speed-dependent state feedback. The longitudinal speed V_x is selected as the scheduling variable because, in the modified bicycle model used to derive the LQR, the system matrices depend explicitly on V_x, which directly changes the lateral tire-force terms, yaw dynamics, and the overall lateral-roll coupling.

The weighting matrices were selected following a normalized design logic similar to Bryson's rule, in which each state and input is penalized according to an admissible maximum value. For the state vector $x = [v\ \dot{\psi}\ \dot{\phi}\ \phi]^\top$, the adopted limits were $v_{\max} = 0.5\,\mathrm{m/s}$, $\dot{\psi}_{\max} = 0.3\,\mathrm{rad/s}$, $\dot{\phi}_{\max} = 0.2\,\mathrm{rad/s}$, and $\phi_{\max} = 0.08\,\mathrm{rad}$, leading to $Q = \mathrm{diag}(1/v_{\max}^2,\ 1/\dot{\psi}_{\max}^2,\ 1/\dot{\phi}_{\max}^2,\ 1/\phi_{\max}^2)$. For the input vector $u = [\delta\ M_z\ M_x]^\top$, the admissible values were taken as $\delta_{\max} = 0.1\,\mathrm{rad}$, $M_{z,\max} = 8000\,\mathrm{Nm}$, and $M_{x,\max} = 1500\,\mathrm{Nm}$, so that $R = 0.1\,\mathrm{diag}(1/\delta_{\max}^2,\ 1/M_{z,\max}^2,\ 1/M_{x,\max}^2)$. The scalar factor 0.1 was introduced as a global aggressiveness coefficient to allow a more responsive controller without over-penalizing the control action. For each scheduling point, the descriptor model is converted to the standard form $\dot{x} = A_{\mathrm{sys}}x + B_{\mathrm{sys}}u$ with $A_{\mathrm{sys}} = -E^{-1}F$ and $B_{\mathrm{sys}} = E^{-1}G$, and the corresponding feedback gain is then obtained from $K_{\mathrm{LQR}} = \mathrm{lqr}(A_{\mathrm{sys}}, B_{\mathrm{sys}}, Q, R)$.

This controller represents a more capable baseline than the fixed-parameter 2PID+fuzzy controller, as it exploits a linearized vehicle model and schedules the state-feedback gains as a function of the longitudinal speed V_x. Its inclusion allows the comparison to contrast the MFC's online disturbance estimation against a structured model-based benchmark with speed-dependent state feedback.

3.3 CarSim Built-in Baseline Controller

The third benchmark is the ESC baseline controller embedded in CarSim®, which provides a representative Electronic Stability Control example for FMVSS 126 / ECE R13H sine-with-dwell testing (Mechanical Simulation Corporation, 2025b).

As described in the official CarSim® FMVSS 126 application example, this built-in

baseline is intended for ESC evaluation under the sine-with-dwell procedure and is assessed through lateral-stability criteria based mainly on yaw-rate decay after the steering input, together with the prescribed lateral-displacement check for high-gain cases (Mechanical Simulation Corporation, 2025b). The public description does not indicate any dedicated roll-control channel; therefore, this baseline is interpreted here as an ESC-oriented lateral-stability controller rather than as a full unified chassis controller with explicit roll regulation.

The public CarSim® description documents the test procedure and the example workflow, but it does not disclose the internal implementation of the embedded ESC in sufficient detail to establish whether the controller uses additional simulator-internal variables beyond the documented test signals (Mechanical Simulation Corporation, 2025b). For that reason, the CarSim® baseline should be viewed as a simulator-native black-box reference. This makes it valuable as a practical benchmark, but also means that its informational symmetry with the other controllers cannot be fully verified from the publicly available documentation.

This baseline is particularly relevant because it represents an industry-relevant reference implemented within the same simulation environment, providing a practical indication of the performance margin achievable with a conventional ESC-oriented strategy under identical test conditions.

4 Model-Free Unified Chassis Control Design

This section presents the proposed MFC-based UCC architecture. The formulation begins with the ultra-local model and the intelligent PID control law, followed by the algebraic estimators for the lumped disturbance term, the overall UCC architecture, and the coordination and control allocation strategies. The section concludes with the parameter selection procedure.

4.1 Ultra-Local Model and Intelligent PID Formulation

The MFC framework adopted in this work is based on the ultra-local model introduced by Fliess and Join (2008), which assumes that the plant input-output behavior can be approximated over a short time window as:

$$y^{(v)} = F + \alpha u, \tag{7}$$

where $y^{(v)}$ denotes the v-th time derivative of the output, F is a lumped term that captures the unknown plant dynamics, external disturbances, and unmodeled nonlinearities, $\alpha \in \mathbb{R}$ is a non-physical scaling parameter chosen by the designer, and u is the control input (Fliess, 2009). The term F is treated as a piecewise-constant function updated at each estimation window, making the representation independent of any explicit vehicle model.

In this work, three plant variables are regulated: the yaw rate $\dot{\psi}$, the side-slip angle β, and the roll angle ϕ. The yaw rate channel is represented by a first-order ultra-local model ($v = 1$), while the side-slip angle and roll angle channels use a second-order formulation ($v = 2$). This choice is consistent with the order of the corresponding vehicle equations: $\dot{\psi}$ is taken as the derivative of the yaw motion, whose dynamics stem from a second-order equation and are therefore adequately captured with $v = 1$, whereas β and ϕ are governed by first-order equations, making a second-order ultra-local description more suitable for their regulation.

Assuming that the ultra-local model holds over a short time window, the stabilizing intelligent PID (iPID) control law is given by:

$$u = \frac{-F + \ddot{y}^*}{\alpha} + K_P e + K_I \int e\, dt + K_D \frac{de}{dt}, \tag{8}$$

where y^* is the reference trajectory, $e = y - y^*$ is the tracking error, and K_P, K_I, K_D are the PID gains. In this work, the integral term is set to zero ($K_I = 0$), reducing the controller to an intelligent PD (iPD) structure. The term $-F/\alpha$ acts as an online feedforward compensation that cancels the estimated lumped dynamics, while the PD component handles residual tracking errors. This choice is appropriate for the SWD maneuver because the control task is dominated by transient stabilization rather than by the elimination of steady-state error.

4.2 Algebraic Estimation of the Lumped Term

The real-time estimation of F is performed using an algebraic approach derived from the ultra-local model via Laplace-domain manipulations, following Fliess (2009). The estimators are expressed as weighted integrals of the input and output signals over a sliding window of length T, which provides inherent noise attenuation without requiring explicit filtering.

4.2.1 Second-order estimator for β and ϕ

For the second-order channels ($v = 2$), the lumped term is estimated as:

$$F_2 = \frac{60}{t^5}\int_0^t (t^2 - 6t\tau + 6\tau^2)y(\tau)\,d\tau - \frac{30\alpha}{t^5}\int_0^t (t-\tau)^2\tau^2 u(\tau)\,d\tau. \quad (9)$$

4.2.2 First-order estimator for $\dot{\psi}$

For the first-order channel ($v = 1$), the estimator takes the form:

$$F_1 = -\frac{6}{t^3}\int_0^t (t - 2\tau)y(\tau)\,d\tau - \frac{6\alpha}{t^3}\int_0^t (t-\tau)\tau\, u(\tau)\,d\tau. \quad (10)$$

4.2.3 Discrete-time implementation

For digital implementation, the integrals in Eqs. (9) and (10) are approximated using the trapezoidal rule over a window of length T, sampled at period T_s. The discrete estimators are:

$$\hat{F}_2 = \frac{60}{T^5}\sum_{k=0}^{T/T_s} a_k(T^2 - 6TkT_s + 6(kT_s)^2)y[k] - \frac{30\hat{\alpha}}{T^5}\sum_{k=0}^{T/T_s} a_k(T-kT_s)^2(kT_s)^2 u[k], \quad (11)$$

$$\hat{F}_1 = -\frac{6}{T^3}\sum_{k=0}^{T/T_s} a_k(T - 2kT_s)y[k] - \frac{6\hat{\alpha}}{T^3}\sum_{k=0}^{T/T_s} a_k(T-kT_s)kT_s\, u[k], \quad (12)$$

where the trapezoidal weights are:

$$a_k = \begin{cases} T_s/2, & k = 0 \text{ and } k = T/T_s, \\ T_s, & k = 1, \ldots, T/T_s - 1. \end{cases} \quad (13)$$

The integration window was set to $T = 1.0\,\mathrm{s}$ for all three estimators, based on the characteristic response time of the vehicle dynamics under the SWD maneuver. This value represents a compromise between estimation smoothness and responsiveness: longer windows improve the accuracy and noise attenuation of $\hat{F}$ but slow down the estimator, whereas shorter windows yield faster convergence at the expense of greater sensitivity to measurement noise; in the present SWD conditions, $T = 1.0\,\mathrm{s}$ provided adequate filtering without introducing excessive delay in the transient control action.

4.3 Unified Chassis Control Architecture

An upstream Multi-layer Coordination (MuC) architecture is adopted for the UCC, as illustrated in Fig. 1. The architecture comprises four layers: supervision, high-level control, coordination, and control allocation. The inputs to the system, namely longitudinal speed V_x, front wheel steering angle δ_f, lateral speed V_y, yaw rate $\dot{\psi}$, roll angle ϕ, and roll rate $\dot{\phi}$, are obtained directly from CarSim®.

4.3.1 Supervision layer

The supervision layer computes the references and defines the activation conditions for each controller channel, as formalized in Section 2.2. The β controller is activated only when $|\beta| > 2°$; the $\dot{\psi}$ reference is computed dynamically via Eq. (3); and the ϕ reference is always zero.

4.3.2 High-level controller

The high-level controller applies three independent iPD controllers, one per regulated variable, to compute the virtual control moments. The lateral dynamics channels (β and $\dot{\psi}$) produce a combined yaw moment M_z, while the roll channel

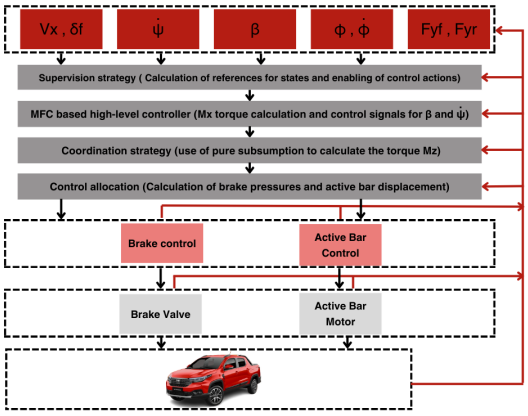


Fig. 1: UCC architecture

produces a roll moment M_x. Both are subject to the saturation limits defined in Eqs. (5)–(6). The implementation in MATLAB/Simulink® is shown in Fig. 2.

4.3.3 Coordination layer

Since both the β and $\dot{\psi}$ controllers contribute to M_z, a Pure Subsumption strategy is used in the coordination layer to weight their respective contributions in the final torque signal. The roll channel is independent and not affected by this coordination step.

4.4 Control Allocation

The virtual control moments M_z and M_x are translated into physical actuator commands by CarSim®. Specifically, M_z is decomposed into individual wheel brake pressure signals processed by the ABS subsystem, and M_x is converted into active anti-roll bar force commands. This allocation step is handled internally by the simulator, consistent with the setup described in (Arronte et al., 2023). Public CarSim® documentation states that the built-in ESC modulates the brakes of individual wheels, but it does not provide the detailed internal allocation law used to map M_z into wheel-level brake pressures, nor an explicit public description of the internal AARB mapping from M_x to actuator commands (Mechanical Simulation Corporation, 2025a,b). Accordingly, in the present work the allocation stage is treated as transparent to the high-level controller: the controller supplies the virtual moments M_z and M_x, while the actuator-level distribution is resolved internally by CarSim®.

4.5 Parameter Selection and Implementation Details

The iPD controllers involve three sets of parameters per channel: the scaling constant α, the proportional gain K_P, and the derivative gain K_D. These were selected empirically through iterative simulation, guided by the following criteria: α was adjusted to match the order of magnitude of the unknown plant gain for each channel, while K_P and K_D were tuned to balance tracking performance and control signal smoothness, prioritizing maneuver safety over passenger comfort.

The effect of α on closed-loop behavior was observed to be monotonic: higher values introduce response delay, while lower values increase control signal oscillation. An intermediate value was sought for each channel independently. It was further verified through simulation that variations in K_D had negligible influence on the system response; therefore, K_D was fixed at a small value for all channels to provide light damping. Although the tuning was empirical, it was guided by safety-oriented criteria in the SWD maneuver, namely preserving closed-loop stability, limiting actuator saturation, and achieving adequate transient regulation, whose effectiveness is confirmed by the results in Section 6.

The resulting parameter values are summarized in Table 1.

Table 1: Selected iPD parameters for each regulated channel

Channel	α	K_P	K_D	Ultra-local model (v)
β	10^{-3}	20	3	2
$\dot{\psi}$	10^{-3}	50	3	1
ϕ	10^{-2}	200	3	2

The simulation sampling period was set to $T_s = 1\,\text{ms}$, and the estimation window to $T = 1.0\,\text{s}$ for all three channels.

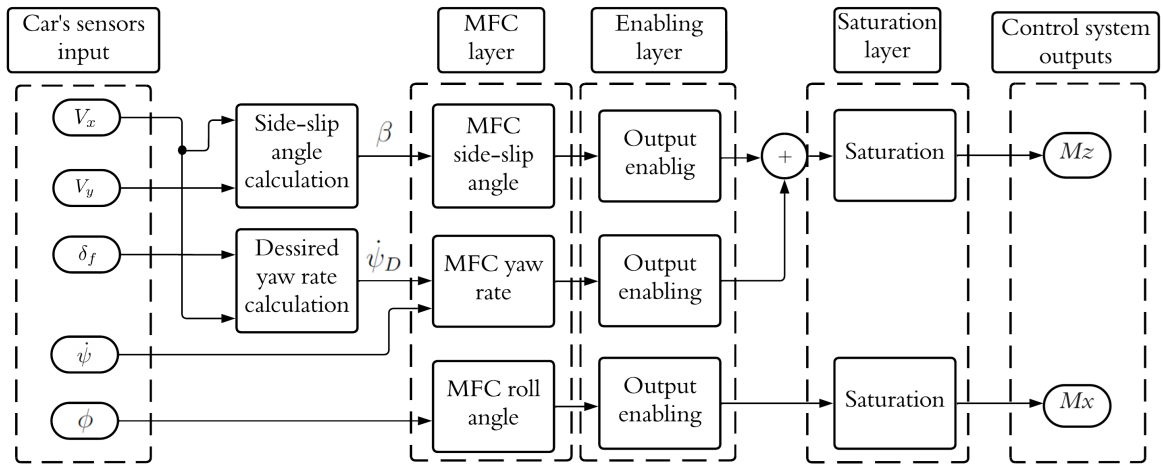


Fig. 2: UCC implementation

5 Experimental Design

To validate the proposed UCC, the closed-loop system is simulated in MATLAB/Simulink® coupled with the CarSim® SUV model parameterized according to Arronte et al. (2023). The experiments focus on the comparison between the proposed iPD-based MFC controller, the gain-scheduled LQR benchmark, and the open-loop vehicle response under the same test conditions, using the SWD maneuver as the common evaluation scenario. To assess robustness, three friction levels are considered (Ahrenhold et al., 2023): $\mu = 0.7$ (dry asphalt), $\mu = 0.5$ (wet road), and $\mu = 0.3$ (icy road).

The SWD maneuver involves accelerating the vehicle to a speed slightly above 80 km/h without steering or braking, maintaining the highest gear up to 80 km/h, and then applying a steering input in the form of a sine wave with a dwell, that is, a short hold at the maximum steering angle. The purpose of the test is to evaluate the vehicle's response to a sudden evasive maneuver, such as swerving to avoid an obstacle, in accordance with regulations such as FMVSS 126 and UN/ECE Regulation No. 13-H Mechanical Simulation Corporation (2012). Figure 3 shows the steering waveform used in this study.

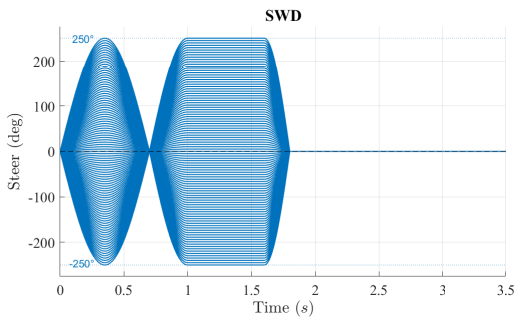


Fig. 3: Steering angle for the sine-with-dwell maneuver

6 Simulation Results

The UCC is evaluated through the SWD maneuver described above under three different road-friction conditions. The results reported below compare the open-loop vehicle response, the CarSim® ESC controller, the fixed-parameter 2PID+fuzzy controller, the gain-scheduled LQR benchmark, and the proposed iPD-based MFC controller.

6.1 Simulation for $\mu = 0.7$

Figures 4a and 4b show the side-slip angle and yaw rate during the SWD maneuver for $\mu = 0.7$. Under this condition, both controllers, namely the gain-scheduled LQR and the iPD-based MFC, reduce the side-slip angle and keep the measured yaw rate close to its reference. By contrast, the vehicle fails to complete the maneuver in open loop.

Figures 4c and 4d show the roll angle and roll rate responses, indicating that both controllers mitigate the rollover-related dynamics during the maneuver.

6.2 Simulation for $\mu = 0.5$

Figures 5a and 5b show the side-slip angle and yaw rate, respectively, during the SWD maneuver with $\mu = 0.5$. Both controllers reduce the side-slip angle and keep the yaw rate close to its reference. As in the previous case, the open-loop vehicle does not complete the maneuver. Figures 5c and 5d show the corresponding roll angle and roll rate responses, which again confirm the beneficial effect of the controllers on rollover-related dynamics.

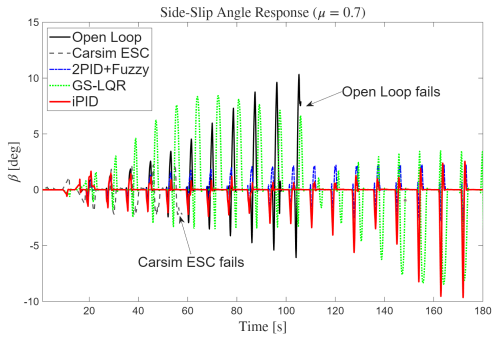
6.3 Simulation for $\mu = 0.3$

Figures 6a and 6b show the side-slip angle and yaw rate, respectively, during the SWD maneuver with $\mu = 0.3$. Under this low-friction condition, only the iPD-based MFC controller is able to complete the maneuver while reducing the side-slip angle and yaw-rate tracking error. Both the open-loop case and the gain-scheduled LQR benchmark fail shortly after the maneuver begins. Figures 6c and 6d show that the same pattern is reflected in the roll dynamics, whereas the proposed MFC controller maintains the roll response within an acceptable range.

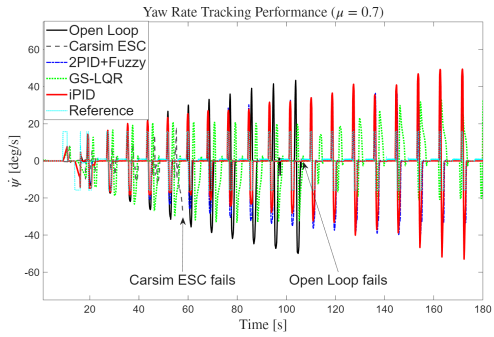
6.4 Statistical results across different friction conditions

Table **??** presents a statistical analysis of the vehicle's dynamic response under the SWD maneuver for the same three road friction coefficients, namely $\mu = 0.7$, $\mu = 0.5$, and $\mu = 0.3$. The results are reported for the two control strategies, gain-scheduled LQR and iPD MFC, as well as for the open-loop case. Cases in which stability is not guaranteed are highlighted in **bold**.

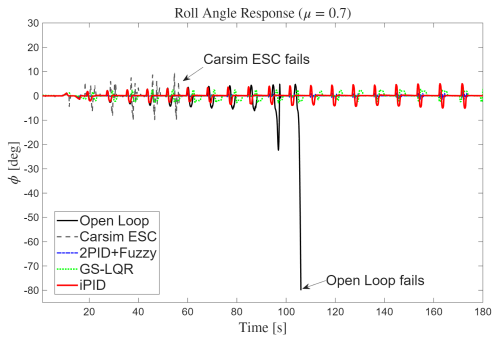
Side-slip angle: Both controllers demonstrate a clear ability to reduce the side-slip angle. The MFC consistently achieves the lowest RMS and RMSE values, particularly at $\mu = 0.5$ and $\mu = 0.3$, where the gain-scheduled LQR benchmark shows slightly higher slip levels. At $\mu = 0.7$, MFC manages a lower RMS (0.627) compared to LQR (0.735), and at $\mu = 0.5$, this difference



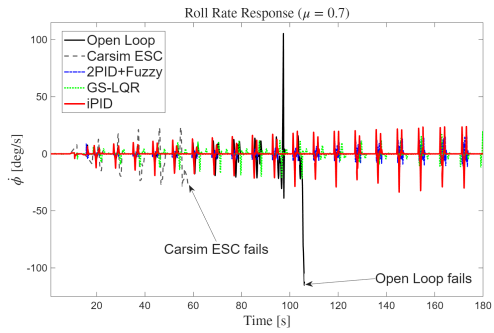
(a) Side-slip angle



(b) Yaw rate

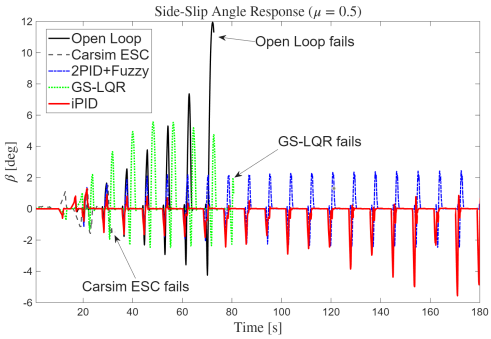


(c) Roll angle

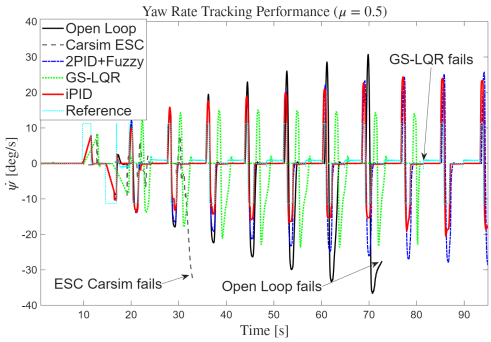


(d) Roll rate

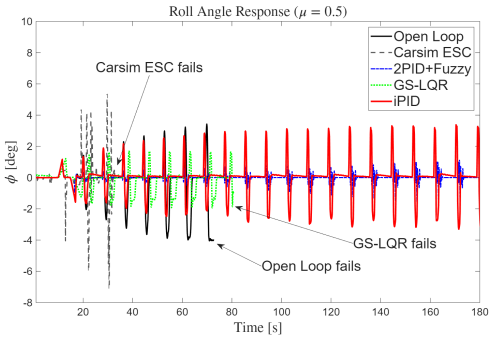
Fig. 4: Results for $\mu = 0.7$



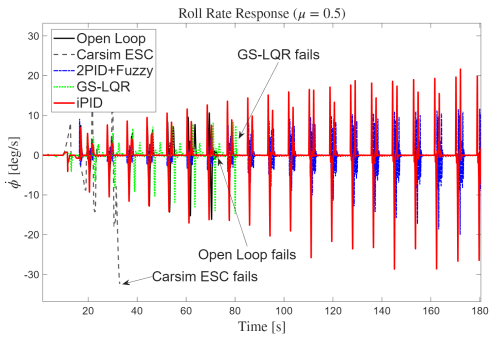
(a) Side-slip angle



(b) Yaw rate

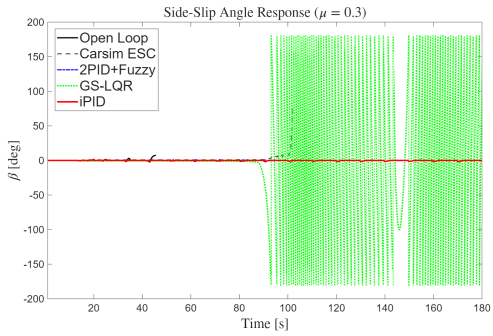


(c) Roll angle

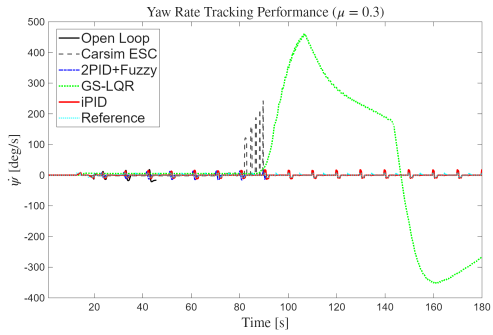


(d) Roll rate

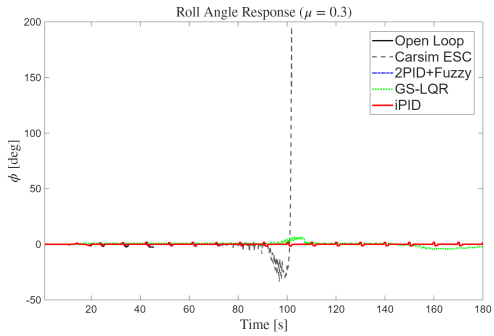
Fig. 5: Results for $\mu = 0.5$



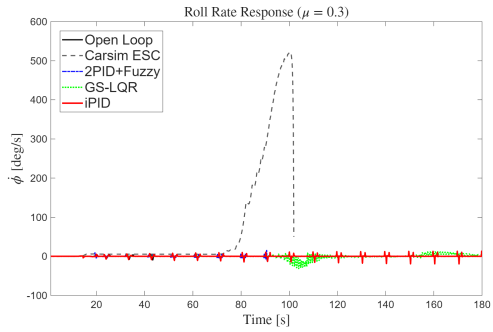
(a) Side-slip angle



(b) Yaw rate



(c) Roll angle



(d) Roll rate

Fig. 6: Results for $\mu = 0.3$

becomes more significant (MFC: 0.531 vs. LQR: 0.71). In the open-loop case, the side-slip angle increases substantially, peaking at 11.945° for $\mu = 0.5$, highlighting the importance of active control for maintaining lateral stability.

Yaw rate: In all conditions, the MFC achieves the lowest yaw rate RMS and RMSE values, followed by the gain-scheduled LQR benchmark. At $\mu = 0.7$, MFC maintains an RMS of 10.770 deg/s and a maximum yaw rate of 34.203 deg/s, while the uncontrolled vehicle reaches a peak of 49.98 deg/s. This pattern continues for $\mu = 0.5$, where MFC maintains better tracking and dampening of yaw motion.

Roll angle: Roll behavior also improves with both controllers. At $\mu = 0.7$, the uncontrolled vehicle experiences an extreme peak roll angle of 79.993, while MFC and the gain-scheduled LQR limit this to 5.917° and 4.161°, respectively. At $\mu = 0.5$, the LQR benchmark shows superior performance in minimizing the roll angle (RMS: 0.222), while MFC reaches a higher RMS (1.260) but still provides acceptable control.

Roll rate: Similar to the roll angle, the roll rate is substantially reduced by the control strategies. At $\mu = 0.7$, the uncontrolled vehicle reaches a peak roll rate of 115.104 deg/s, whereas the LQR and MFC controllers keep it below 42 deg/s. The same trend is observed for $\mu = 0.5$, with the MFC showing higher values than LQR at $\mu = 0.5$.

For $\mu = 0.3$, the gain-scheduled LQR benchmark and the open-loop case do not remain stable. Overall, the analysis demonstrates that both control strategies significantly enhance vehicle stability and rollover mitigation during the SWD maneuver for $\mu = 0.7$ and $\mu = 0.5$, with the MFC generally offering superior performance in minimizing side-slip and yaw-rate errors. The roll angle and roll rate are better controlled by the LQR benchmark. On the other hand, the iPD-based MFC exhibits greater stability robustness, since it does not fail when $\mu = 0.3$. The uncontrolled vehicle is not robustly stable.

7 Conclusions

This study proposes a novel Unified Chassis Control (UCC) system based on Model-Free Controllers (MFCs) to simultaneously regulate vehicle lateral (side-slip angle and yaw rate) and vertical (roll angle) dynamics during abrupt maneuvers. By leveraging ultra-local models and algebraic estimators, the MFC-based UCC eliminates the need for detailed vehicle modeling, effectively addressing challenges posed by parametric uncertainties and the nonlinear nature of traditional model-based approaches. The controller was evaluated through an SWD maneuver in a co-simulation environment using MATLAB/Simulink® and CarSim®, under varying road friction conditions ($\mu = 0.7$, $\mu = 0.5$, and $\mu = 0.3$), and compared against a gain-scheduled LQR benchmark. Results showed that the MFC-based UCC significantly reduced side-slip angle and yaw-rate errors while maintaining good roll-angle stability, ensuring comprehensive lateral and vertical control. In low-friction scenarios ($\mu = 0.3$), only the MFC remained stable.

Future research should focus on automating the tuning using optimization to enhance scalability and precision, validating the approach through real-world implementations, and comparing it with alternative model-free techniques such as neural networks or fuzzy logic controllers. Additionally, extending the UCC framework to account for longitudinal dynamics or adapting it for diverse vehicle types, including electric and heavy-duty platforms, could further expand its applicability.

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Parameter	μ	Metric	Open Loop	Carsim® ESC	2PID+Fuzzy	GS-LQR	iPID
β	0.3	RMS	1.3865	4.3328	0.2802	84.0626	0.3820
		RMSE	1.3865	4.3328	0.2802	84.0626	0.3820
		MAX	7.4658	79.9872	1.5401	180	2.0773
		ST	45.6040	101.8755	90.8950	180.0	180.0
	0.5	RMS	2.0735	0.6085	0.7095	1.4398	0.7330
		RMSE	2.0735	0.6085	0.7095	1.4398	0.7330
		MAX	11.9446	1.6017	2.5307	5.5587	5.7423
		ST	72.7040	32.8805	180.0	80.7220	180.0
	0.7	RMS	1.8257	0.8429	0.6793	2.4362	1.0634
		RMSE	1.8257	0.8429	0.6793	2.4362	1.0634
		MAX	10.3295	2.2731	2.8959	8.4473	10.6926
		ST	106.0340	57.8050	180.0	180.0	180.0
$\dot{\psi}$	0.3	RMS	6.2029	112.9537	4.2652	161.4949	4.7386
		RMSE	4.7779	112.8709	2.0899	11025	2.7861
		MAX	21.3541	444.0231	15.6547	559.4037	17.2675
		ST	45.6040	101.8755	90.8950	180.0	180.0
	0.5	RMS	10.3026	6.8931	9.8975	6.3843	8.9371
		RMSE	7.7189	6.8115	5.1593	12.3886	5.2175
		MAX	36.6287	22.2112	30.4512	23.6014	38.6147
		ST	72.7040	32.8805	180.0	80.7220	180.0
	0.7	RMS	12.7302	8.5793	12.0261	9.5523	12.1507
		RMSE	8.6956	8.4799	8.9794	17.7459	7.4796
		MAX	49.9795	30.1345	41.4211	33.1855	56.7753
		ST	106.0340	57.8050	180.0	180.0	180.0
ϕ	0.3	RMS	0.8038	10.0893	0.1644	5.3870	0.5981
		RMSE	0.8038	10.0893	0.1644	5.3870	0.5981
		MAX	2.4516	195.4223	1.1661	77.7313	1.8932
		ST	45.6040	101.8755	90.8950	180.0	180.0
	0.5	RMS	1.3398	1.2443	0.2223	0.7291	1.0816
		RMSE	1.3398	1.2443	0.2223	0.7291	1.0816
		MAX	4.0499	7.1238	1.8518	1.9372	3.3769
		ST	72.7040	32.8805	180.0	80.7220	180.0
	0.7	RMS	4.6383	1.9532	0.2275	1.0800	1.4269
		RMSE	4.6383	1.9532	0.2275	1.0800	1.4269
		MAX	79.9933	10.1961	1.8189	2.8552	5.1379
		ST	106.0340	57.8050	180.0	180.0	180.0
$\dot{\phi}$	0.3	RMS	1.5275	161.6824	1.1813	23.3285	2.3198
		RMSE	1.5275	161.6824	1.1813	23.3285	2.3198
		MAX	9.4818	522.0664	15.4329	583.2962	22.3129
		ST	45.6040	101.8755	90.8950	180.0	180.0
	0.5	RMS	2.7989	6.7776	2.4884	2.0264	3.7970
		RMSE	2.7989	6.7776	2.4884	2.0264	3.7970
		MAX	16.1562	32.3786	20.6892	14.6179	31.4270
		ST	72.7040	32.8805	180.0	80.7220	180.0
	0.7	RMS	9.8414	7.9221	2.4859	2.9383	4.6610
		RMSE	9.8414	7.9221	2.4859	2.9383	4.6610
		MAX	115.1039	28.9968	18.1210	21.3752	33.4922
		ST	106.0340	57.8050	180.0	180.0	180.0

Table 2: Performance comparison of control strategies under varying friction coefficients.

Parameter	μ	Metric	Carsim® ESC	2PID+Fuzzy	GS-LQR	iPID
		RMS	23.4196	2.6088	2.2192	8.2063
		MAX	55.2815	11.8114	1.68E+03	23.9273
	0.3	Energy	55876	618.5951	1.77E+03	2.42E+04
		Sat	NA	0	0.01	6.7227
		Rate RMS	557.7514	218.6228	3.14E+03	7.83E+03
		RMS	2.8170	4.9124	0.3941	6.5678
		MAX	12	66.7595	96.0468	23.9273
Pbs	0.5	Energy	260.9212	8.51E+03	12.5355	1.51E+04
		Sat	NA	1.5661	0.0056	1.6008
		Rate RMS	93.5817	861.0666	557.3531	5.97E+03
		RMS	3.4939	5.0176	0.7799	7.3820
		MAX	12	22.9088	405.6450	23.9273
	0.7	Energy	705.6629	8.94E+03	121.6541	1.93E+04
		Sat	NA	3.7111	0.0037	2.9302
		Rate RMS	151.9735	572.9802	1.10E+03	9.06E+03
		RMS	492.0474	897.7965	1.55E+03	206.0696
		MAX	2228	3.39E+03	1.09E+04	962.1785
	0.3	Energy	2.13E+10	7.33E+07	8.67E+08	1.53E+07
		Sat	NA	12.5566	22.0652	0
		Rate RMS	764.1464	5.93E+03	9.26E+03	3.95E+03
		RMS	210.4468	1.98E+03	148.8898	351.4349
		MAX	525.3417	6.74E+03	669.5755	1200
Mx	0.5	Energy	1.46E+06	1.38E+09	1.79E+06	4.34E+07
		Sat	NA	22.9564	0	0.2259
		Rate RMS	386.8675	9.40E+03	617.4531	7.32E+03
		RMS	271.0004	2.42E+03	243.2953	455.9170
		MAX	731.5411	9.18E+03	835.7442	1200
	0.7	Energy	2.94E+06	2.07E+09	1.18E+07	7.37E+07
		Sat	NA	22.3799	0	4.52
		Rate RMS	539.0644	9.85E+03	841.1845	8.04E+03

Table 3: Control effort comparison in terms of brake pressure (Pbs) and anti-roll torque (Mx).

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